

CSE321

Embedded Systems Miniproject Report

Vibhaas Nirantar Srivastava
2023BCS0037

Machine Temperature Monitoring and Overheat Alert System

Project Description

Industrial machines are prone to overheating, which can lead to failure, downtime, and hazards. In this project we design an automated temperature monitoring system using an ARM7 microcontroller (LPC2148). The system continuously reads the machine's temperature via an LM35 sensor, displays the real-time data on an LCD, and triggers a visual alert if the temperature exceeds a safe predefined threshold (45°C).

Hardware Components & Architecture

1. Microcontroller (LPC2148): Acts as the brain of the system, featuring an onboard 10-bit Analog-to-Digital Converter (ADC) necessary for reading analog sensor data.
2. LM35 Temperature Sensor: An integrated-circuit temperature device with an output voltage linearly proportional to the Centigrade temperature (10mV/°C).
3. 16x2 Alphanumeric LCD: Used for the real-time temperature readings.
4. Alert Mechanism (LED): Provides immediate warning when the overheat condition is met.

System Implementation

- **Input Stage:** The LM35 is connected to pin P0.28, which is configured as ADC channel AD0.1. The LPC2148 converts the analog voltage into a digital value ranging from 0 to 1023.
 - **Processing Stage:** The digital value is passed through a conversion formula:
$$\text{Temperature} = (\text{ADC_Value} * 3.3\text{V} * 100) / 1024.$$
-

-
- **Output Stage:** The microcontroller sends the formatted string to the LCD (connected to P1.16 - P1.25). Simultaneously, a conditional loop checks if the temperature is $> 45^{\circ}\text{C}$. If true, P0.10 is pulled HIGH to activate the alert.

Code

// in Keil uVision 4, Embedded C, to be run on Proteus

```
#include <lpc214x.h>
```

```
#include <stdio.h>
```

```
#define THRESHOLD_TEMP 45.0
```

```
#define LCD_RS (1<<24)
```

```
#define LCD_EN (1<<25)
```

```
#define ALERT_PIN (1<<10)
```

```
void delay_ms(unsigned int ms) {
```

```
    unsigned int i, j;
```

```
    for(i=0; i<ms; i++)
```

```
        for(j=0; j<6000; j++);
```

```
}
```

```
void lcd_cmd(unsigned char cmd) {
```

```
    IO1CLR = 0x03FF0000;
```

```
IO1SET = (cmd << 16);

IO1CLR = LCD_RS;

IO1SET = LCD_EN;

delay_ms(2);

IO1CLR = LCD_EN;

}
```

```
void lcd_data(unsigned char data) {

    IO1CLR = 0x03FF0000;

    IO1SET = (data << 16);

    IO1SET = LCD_RS;

    IO1SET = LCD_EN;

    delay_ms(2);

    IO1CLR = LCD_EN;

}
```

```
void lcd_init() {

    IO1DIR |= 0x03FF0000;

    lcd_cmd(0x38);

    lcd_cmd(0x0C);

    lcd_cmd(0x01);

}
```

```
    lcd_cmd(0x80);
```

```
}
```

```
void lcd_string(char *str) {
```

```
    while(*str) {
```

```
        lcd_data(*str++);
```

```
    }
```

```
}
```

```
unsigned int adc_read() {
```

```
    unsigned int val;
```

```
    PINSEL1 |= (1<<24);
```

```
    PINSEL1 &= ~(1<<25);
```

```
    AD0CR = 0x01200002;
```

```
    while(!(AD0GDR & 0x80000000));
```

```
    val = (AD0GDR >> 6) & 0x3FF;
```

```
    return val;
```

```
}
```

```
int main() {  
  
    unsigned int adc_val;  
  
    float temp;  
  
    char buffer[16];  
  
    int temp_int, temp_frac;  
  
  
    IOODIR |= ALERT_PIN;  
  
    IOOCLR = ALERT_PIN;  
  
  
    lcd_init();  
  
    lcd_string("Machine Temp:");  
  
    delay_ms(1000);  
  
  
    while(1) {  
  
        adc_val = adc_read();  
  
  
  
        temp = ((float)adc_val * 3.3 * 100.0) / 1024.0;  
  
  
  
        temp_int = (int)temp;  
  
        temp_frac = (int)((temp - temp_int) * 100);  
  
    }  
}
```

```
    sprintf(buffer, "Temp: %d.%02d C ", temp_int, temp_frac);

    lcd_cmd(0x80);

    lcd_string(buffer);

    lcd_cmd(0xC0);

    if (temp > THRESHOLD_TEMP) {

        IO0SET = ALERT_PIN;

        lcd_string("!! OVERHEAT !! ");

    } else {

        IO0CLR = ALERT_PIN;

        lcd_string("Status: NORMAL ");

    }

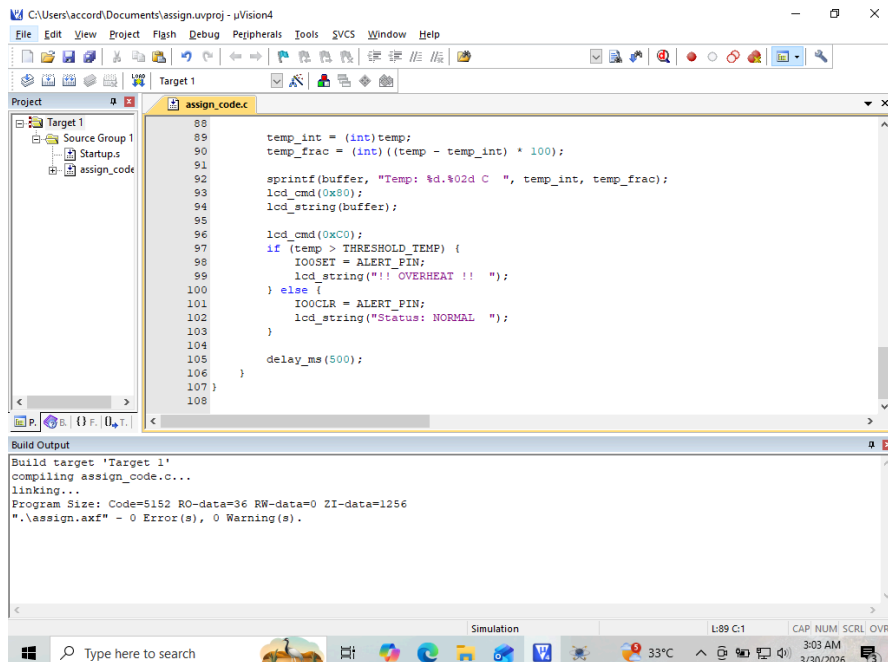
    delay_ms(500);

}

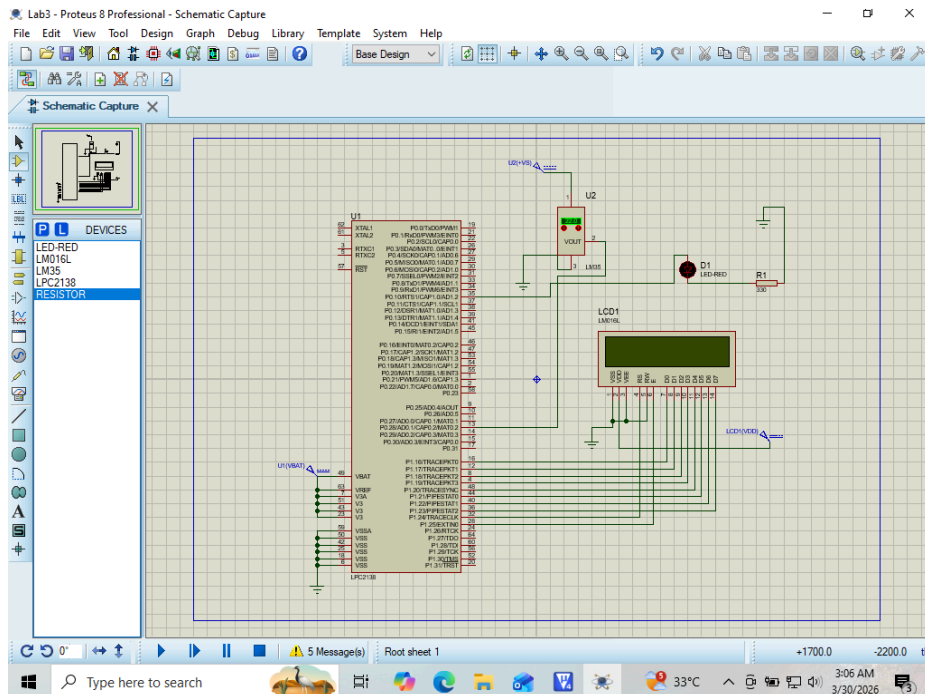
}
```

Simulation

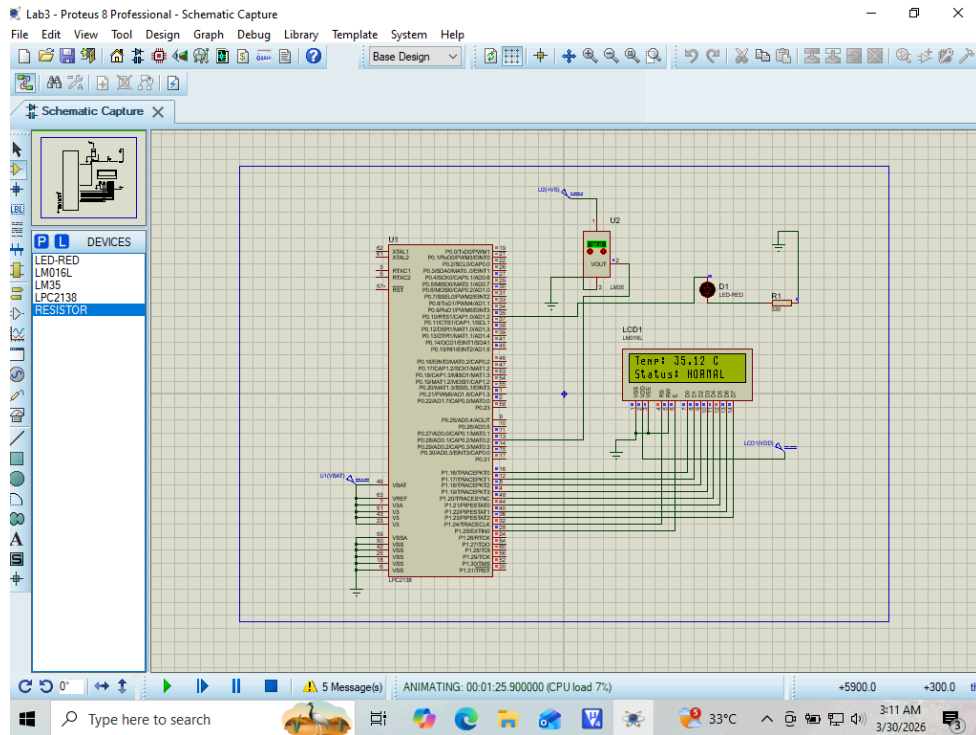
- Code compilation on Keil :



- Setting up the connections:



- Example of normal temperature execution:



- Example of overheat:

